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Previous Year Paper (Mechanical) 19 Jan, 2025 Shift 1

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QUESTION PAPER
Mechanical [SHIFT - 1]

Exam Date : 19/01/2025

Time : 03:00 PM - 05:00 PM

Part I - Aptitude

Question 1

Question Id : 376

Directions: *Study the passage below to answer the question that follows.*

Passage
To those who do listen, the desert speaks of things with an emphasis quite different from that of the shore, the mountain, the valley or the plains. Whereas these invite action and suggest limitless opportunity and exhaustless resources, the implications and the mood of the desert are something different. For one thing, the desert is conservative, not radical. It is more likely to provide awe than to invite conquest. The heroism which it encourages is the heroism of endurance, not that of conquest. It brings man up against this limitation, turns him upon himself and suggests values which more indulgent regions suppress. Sometimes it induces contemplation in men who have never contemplated before. And of all the answers to the question—what is a desert good for—'contemplation' is perhaps the best.

In order to receive the desert's message, the beholder needs to be

Answer :

Option Id

(A) courageous in his reaction. 376001

(B) conservative in his responses. 376002

(C) a good listener. 376003

(D) sensitive to nature. 376004

Right Answer :
sensitive to nature.

Right Option Id : 376004

Question 2

Question Id : 377

Directions: *Study the passage below to answer the question that follows.*

Passage
To those who do listen, the desert speaks of things with an emphasis quite different from that of the shore, the mountain, the valley or the plains. Whereas these invite action and suggest limitless opportunity and exhaustless resources, the implications and the mood of the desert are something different. For one thing, the desert is conservative, not radical. It is more likely to provide awe than to invite conquest. The heroism which it encourages is the heroism of endurance, not that of conquest. It brings man up against this limitation, turns him upon himself and suggests values which more indulgent regions suppress. Sometimes it induces contemplation in men who have never contemplated before. And of all the answers to the question—what is a desert good for—'contemplation' is perhaps the best.

The desert is unique among landscapes in that it encourages

Answer :

Option Id

(A) contemplation. 377001

(B) indolence. 377002

(C) heroic endeavor. 377003

(D) adventurous spirit. 377004

Right Answer :
contemplation.

Right Option Id : 377001

Question 3

Question Id : 378

Directions: *Study the passage below to answer the question that follows.*

Passage
To those who do listen, the desert speaks of things with an emphasis quite different from that of the shore, the mountain, the valley or the plains. Whereas these invite action and suggest limitless opportunity and exhaustless resources, the implications and the mood of the desert are something different. For one thing, the desert is conservative, not radical. It is more likely to provide awe than to invite conquest. The heroism which it encourages is the heroism of endurance, not that of conquest. It brings man up against this limitation, turns him upon himself and suggests values which more indulgent regions suppress. Sometimes it induces contemplation in men who have never contemplated before. And of all the answers to the question—what is a desert good for—'contemplation' is perhaps the best.

If one responds with insight to the mood of the desert, it evokes

Answer :

Option Id

(A) the possibility of unending resources. 378001

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(B) an inclination for deep thought.
(C) the desire for heroic conquest.
(D) a sense of intense revulsion.

378003
378004

Right Answer :
an inclination for deep thought.

Right Option Id : 378002

Question 4

Question Id : 379

Directions: *Study the passage below to answer the question that follows.*

Passage
To those who do listen, the desert speaks of things with an emphasis quite different from that of the shore, the mountain, the valley or the plains. Whereas these invite action and suggest limitless opportunity and exhaustless resources, the implications and the mood of the desert are something different. For one thing, the desert is conservative, not radical. It is more likely to provide awe than to invite conquest. The heroism which it encourages is the heroism of endurance, not that of conquest. It brings man up against this limitation, turns him upon himself and suggests values which more indulgent regions suppress. Sometimes it induces contemplation in men who have never contemplated before. And of all the answers to the question—what is a desert good for—'contemplation' is perhaps the best.

The writer calls the desert 'conservative' rather than 'radical' because it provides an environment that

Answer : Option Id
(A) inspires man to explore it. 379001
(B) offers unlimited opportunity to conquer. 379002
(C) tests one's endurance. 379003
(D) makes one gloomy. 379004

Right Answer :
tests one's endurance.

Right Option Id : 379003

Question 5

Question Id : 380

Directions: *Study the passage below to answer the question that follows.*

Passage
To those who do listen, the desert speaks of things with an emphasis quite different from that of the shore, the mountain, the valley or the plains. Whereas these invite action and suggest limitless opportunity and exhaustless resources, the implications and the mood of the desert are something different. For one thing, the desert is conservative, not radical. It is more likely to provide awe than to invite conquest. The heroism which it encourages is the heroism of endurance, not that of conquest. It brings man up against this limitation, turns him upon himself and suggests values which more indulgent regions suppress. Sometimes it induces contemplation in men who have never contemplated before. And of all the answers to the question—what is a desert good for—'contemplation' is perhaps the best.

What does the phrase "it brings man up against his limitations", mean ?

Answer : Option Id
(A) It makes man feel hopeless about his limitations. 380001
(B) It makes man aware of his limitations. 380002
(C) It compels man to fight for his limitations. 380003
(D) It persuades man to overcome his limitations. 380004

Right Answer :
It persuades man to overcome his limitations.

Right Option Id : 380004

Question 6

Question Id : 381

The salary of a school teacher was increased by 25% and later it was decreased by 25%. If he now draws Rs.937.50, what was the original salary?

Answer : Option Id
(A) Rs.1000 381001
(B) Rs.990 381002
(C) Rs.985 381003
(D) Rs.980 381004

Right Answer :
Rs.1000

Right Option Id : 381001

Question 7

Question Id : 382

The average age of 10 men increases by 2 years when one of them who is of 25 years age, is replaced by a new man. The age of the new man is _____.

Answer :

Option Id

(A) 34 years — 382001

(B) 37 years — 382002

(C) 41 years — 382003

(D) 45 years — 382004

Right Answer :

45 years

Right Option Id : 382004

Question 8

Question Id : 383

In how many minimum number of complete years, a sum of money put out at 25 percent compound interest will be more than doubled?

Answer :

Option Id

(A) 3 years — 383001

(B) 6 years — 383002

(C) 4 years — 383003

(D) 9 years — 383004

Right Answer :

4 years

Right Option Id : 383003

Question 9

Question Id : 384

A man and a boy finish a certain piece of work in 20 days whereas a man alone can do it in 25 days. If the man does the work 'x' times as fast as the boy, find 'x'.

Answer :

Option Id

(A) 6 — 384001

(B) 4 — 384002

(C) 7 — 384003

(D) 9 — 384004

Right Answer :

4

Right Option Id : 384002

Question 10

Question Id : 385

In what proportion must sugar at Rs.16.60 per kg be mixed with sugar at Rs.16.45 per kg so that the mixture may be worth Rs.16.54 per kg?

Answer :

Option Id

(A) 2 : 5 — 385001

(B) 2 : 3 — 385002

(C) 3 : 2 — 385003

(D) 3 : 4 — 385004

Right Answer :

3 : 2

Right Option Id : 385003

Question 11

Question Id : 386

Complete the series by replacing '?' with the correct option.

U7Z, V6A, W5B, X4C, Y3D, '?'

Answer :

Option Id

(A) Z1F — 386001

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(B) Z2E
(C) X3F
(D) Y4G

386003
386004

Right Answer :
Z2E

Right Option Id : 386002

Question 12

Question Id : 387

Find the odd one out in the series.

APL, ZQK, YRJ, XSI, WTK, VUG

Answer :

Option Id

(A) ZQK — 387001
(B) YRJ — 387002
(C) WTK — 387003
(D) VUG — 387004

Right Answer :
WTK

Right Option Id : 387003

Question 13

Question Id : 388

L, M, N, O, P and Q are sitting in a row all facing front. P and Q are sitting in the centre. L and M are sitting at the two ends. N is sitting to the right of L. Who is sitting to the immediate left of M?

Answer :

Option Id

(A) O — 388001
(B) L — 388002
(C) P — 388003
(D) Q — 388004

Right Answer :
O

Right Option Id : 388001

Question 14

Question Id : 389

If in a certain code, FLOWERS is written as SREWOLG, how will MOUNTAIN be written in that language?

Answer :

Option Id

(A) NIATNUOM — 389001
(B) NIATNOUN — 389002
(C) NIANTUON — 389003
(D) NIATNUON — 389004

Right Answer :
NIATNUON

Right Option Id : 389004

Question 15

Question Id : 390

Four tickets for movie A and five tickets for movie B together cost Rs.1540 but five tickets for movie A and four tickets for movie B cost Rs.1460. What is the cost of the tickets for movie B?

Answer :

Option Id

(A) Rs.450 — 390001
(B) Rs.400 — 390002
(C) Rs.333.34 — 390003
(D) Rs.206.67 — 390004

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Right Answer :
Rs.206.67

Right Option Id : 390004

Question 16

Question Id : 391

Which one of the following metals is the most abundant in the Earth's crust?

Answer :

Option Id

(A) Zinc — 391001
(B) Copper — 391002
(C) Aluminium — 391003
(D) Iron — 391004

Right Answer :
Aluminium

Right Option Id : 391003

Question 17

Question Id : 392

Which of the following cities is the birthplace of Swami Vivekananda?

Answer :

Option Id

(A) Cuttack — 392001
(B) Balasore — 392002
(C) Kushinagar — 392003
(D) Calcutta — 392004

Right Answer :
Calcutta

Right Option Id : 392004

Question 18

Question Id : 393

The Constitution of India names our country as_____.

Answer :

Option Id

(A) Bharat — 393001
(B) India, that is Bharat — 393002
(C) Hindustan — 393003
(D) Aryavarta — 393004

Right Answer :
India that is Bharat

Right Option Id : 393002

Question 19

Question Id : 394

Where is the headquarters of Asian Development Bank located?

Answer :

Option Id

(A) Philippines — 394001
(B) India — 394002
(C) Japan — 394003
(D) China — 394004

Right Answer :
Philippines

Right Option Id : 394001

Question 20

Question Id : 395

Which among the following planets of our Solar system is the biggest in size?

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Answer :

(A) Venus
(B) Mercury
(C) Earth
(D) Mars

Option Id
395001
395002
395003
395004

Right Answer :
Earth

Right Option Id : 395003

Question 21

Question Id : 396

Directions: *Study the following information and answer the question that follows.*

OrangeCorp Inc. from Florida had acquired recently four ventures in India A, B, C & D. On completion of FY 2023-24, it was found that while the profits of C and D were the same, the sales of C were the same as those of B. Profits of A were 10% of its sales, whereas the profits of B were 20% of its sales. While the total expenses of C were 5 times its profits, the sales of D were 3 times its profits. The total expenses of C were Rs.100 Crore and of A were 10% less than those of C. Profits are the difference between the sales and total expenses.

Which venture had the lowest sales in FY 2023-24?

Answer :

(A) A
(B) B
(C) C
(D) D

Option Id
396001
396002
396003
396004

Right Answer :
D

Right Option Id : 396004

Question 22

Question Id : 397

Directions: *Study the following information and answer the question that follows.*

OrangeCorp Inc. from Florida had acquired recently four ventures in India A, B, C & D. On completion of FY 2023-24, it was found that while the profits of C and D were the same, the sales of C were the same as those of B. Profits of A were 10% of its sales, whereas the profits of B were 20% of its sales. While the total expenses of C were 5 times its profits, the sales of D were 3 times its profits. The total expenses of C were Rs.100 Crore and of A were 10% less than those of C. Profits are the difference between the sales and total expenses.

Which venture had the highest profits?

Answer :

(A) A
(B) B
(C) C
(D) D

Option Id
397001
397002
397003
397004

Right Answer :
B

Right Option Id : 397002

Question 23

Question Id : 398

Directions: *Study the following information and answer the question that follows.*

OrangeCorp Inc. from Florida had acquired recently four ventures in India A, B, C & D. On completion of FY 2023-24, it was found that while the profits of C and D were the same, the sales of C were the same as those of B. Profits of A were 10% of its sales, whereas the profits of B were 20% of its sales. While the total expenses of C were 5 times its profits, the sales of D were 3 times its profits. The total expenses of C were Rs.100 Crore and of A were 10% less than those of C. Profits are the difference between the sales and total expenses.

Which venture had the highest total expenses?

Answer :

(A) A
(B) B
(C) C
(D) D

Option Id
398001
398002
398003
398004

Right Answer :
C

Right Option Id : 398003

Question 24

Question Id : 399

Directions: *Study the following information and answer the question that follows.*

OrangeCorp Inc. from Florida had acquired recently four ventures in India A, B, C & D. On completion of FY 2023-24, it was found that while the profits of C and D were the same, the sales of C were the same as those of B. Profits of A were 10% of its sales, whereas the profits of B were 20% of its sales. While the total expenses of C were 5 times its profits, the sales of D were 3 times its profits. The total expenses of C were Rs.100 Crore and of A were 10% less than those of C. Profits are the difference between the sales and total expenses.

Which venture had the lowest profits?

Answer :

Option Id

(A) A — 399001
(B) B — 399002
(C) C — 399003
(D) D — 399004

Right Answer :
A

Right Option Id : 399001

Question 25

Question Id : 400

Directions: *Study the following information and answer the question that follows.*

OrangeCorp Inc. from Florida had acquired recently four ventures in India A, B, C & D. On completion of FY 2023-24, it was found that while the profits of C and D were the same, the sales of C were the same as those of B. Profits of A were 10% of its sales, whereas the profits of B were 20% of its sales. While the total expenses of C were 5 times its profits, the sales of D were 3 times its profits. The total expenses of C were Rs.100 Crore and of A were 10% less than those of C. Profits are the difference between the sales and total expenses.

Profit of venture B was what percent of venture A?

Answer :

Option Id

(A) 240 — 400001
(B) 2400 — 400002
(C) 24 — 400003
(D) 2.4 — 400004

Right Answer :
240

Right Option Id : 400001

Part II - Technical

Question 26

Question Id : 401

Which tool is mainly used for checking the diameter of shafts having a central hole, and comes in two types: GO and NOGO, for checking the upper and lower limits of the shaft respectively?

Answer :

Option Id

(A) Ring gauges — 401001
(B) Taper plug gauge — 401002
(C) Taper ring gauge — 401003
(D) None of these — 401004

Right Answer :
Ring gauges

Right Option Id : 401001

Question 27

Question Id : 402

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Which system measures internal dimensions without physical contact and can inspect a single or multiple dimensions simultaneously, often used during or immediately after the operating cycle of a machine tool?

Answer :

Option Id

(A) Pneumatic Comparator — 402001
(B) Dial Indicator — 402002
(C) Digital Caliper — 402003
(D) Micrometer — 402004

Right Answer :

Pneumatic Comparator

Right Option Id : 402001

Question 28

Question Id : 403

A technician observes that an inductor in a DC circuit blocks a sudden surge in current. Which property of the inductor is this behaviour due to?

Answer :

Option Id

(A) Induced EMF opposing changes in current — 403001
(B) Energy dissipation in the coil — 403002
(C) Magnetic field stability at high currents — 403003
(D) Low resistance of the winding material — 403004

Right Answer :

Induced EMF opposing changes in current

Right Option Id : 403001

Question 29

Question Id : 404

A diode is reverse-biased in a circuit. What happens to the majority carriers, and how does the current flow behave?

Answer :

Option Id

(A) Majority carriers recombine, and current increases significantly. — 404001
(B) Majority carriers move away from the junction, and the current flow stops. — 404002
(C) Majority carriers move towards the junction and the current decreases. — 404003
(D) The majority of carriers move away from the junction, but a small current flows due to minority carriers. — 404004

Right Answer :

The majority of carriers move away from the junction but a small current flows due to minority carriers.

Right Option Id : 404004

Question 30

Question Id : 405

In a regulated DC power supply circuit, which diode is suitable for voltage regulation, and what feature of this diode enables this function?

Answer :

Option Id

(A) Photo diode, light intensity detection — 405001
(B) LED, forward bias light emission — 405002
(C) Zener diode, reverse bias breakdown — 405003
(D) Zener diode, forward bias light emission — 405004

Right Answer :

Zener diode reverse bias breakdown

Right Option Id : 405003

Question 31

Question Id : 406

A circuit designer needs to create a buffer circuit for impedance matching. Which transistor configuration is most suitable, and why?

Answer :

Option Id

(A) Common Base (CB) configuration, because it provides high voltage gain and medium power gain. — 406001
(B) Common Emitter (CE) configuration, because it offers high current and voltage gain. — 406002
(C) Common Collector (CC) configuration, because it provides high input impedance and low output impedance. — 406003
(D) Common Emitter (CE) configuration, because it provides high thermal stability. — 406004

Right Answer :
Common Collector (CC) configuration because it provides high input impedance and low output impedance.

Right Option Id : 406003

Question 32

Question Id : 407

A circuit designer observes that the output voltage changes significantly when the power supply voltage fluctuates. Which parameter of the Op Amp is likely to be inadequate?

Answer :	Option Id
(A) Common Mode Rejection Ratio (CMRR)	407001
(B) Power Supply Rejection Ratio (PSRR)	407002
(C) Output voltage swing	407003
(D) Input offset voltage drift	407004

Right Answer :
Power Supply Rejection Ratio (PSRR)

Right Option Id : 407002

Question 33

Question Id : 408

A designer is building a modulus-16 counter. How many flip-flops are required?

Answer :	Option Id
(A) 2 flip-flops	408001
(B) 3 flip-flops	408002
(C) 4 flip-flops	408003
(D) 5 flip-flops	408004

Right Answer :
4 flip-flops

Right Option Id : 408003

Question 34

Question Id : 409

A production line frequently experiences overproduction and defects due to inconsistent demand and process variability. Which Lean principle should the manager focus on to address these issues comprehensively?

Answer :	Option Id
(A) Customer pull, to align production with real-time demand.	409001
(B) Level loading, to smooth out process variability and balance workloads.	409002
(C) Waste elimination, to target and minimize specific inefficiencies.	409003
(D) Standard work, to ensure consistent execution across the production line.	409004

Right Answer :
Level loading to smooth out process variability and balance workloads.

Right Option Id : 409002

Question 35

Question Id : 410

A team identifies performance issues in their manufacturing process and has collected data on key metrics. According to DMAIC, what is the next logical step, and why?

Answer :	Option Id
(A) Define, to outline the root cause of the problem.	410001
(B) Analyze, to prioritize potential causes and suggest solutions.	410002
(C) Control, to implement sustainable solutions for enhanced performance.	410003
(D) Improve, to identify and test innovations for targeted improvements.	410004

Right Answer :
Analyze to prioritize potential causes and suggest solutions.

Right Option Id : 410002

Question 36

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After identifying high-cost and low-performance areas in the initial process steps, what is the next action a production engineer should take according to the process and equipment selection methodology?

Answer :

Option Id

(A) Develop alternative processes to address identified inefficiencies. 411001

(B) Perform a financial analysis to determine the most profitable process. 411002

(C) Communicate with all relevant departments to finalize the process selection. 411003

(D) Initiate detailed processing to implement the selected process. 411004

Right Answer :

Develop alternative processes to address identified inefficiencies.

Right Option Id : 411001

Question 37

Question Id : 412

In a flow manufacturing system, work-in-progress (WIP) stock levels remain low. Which factor most directly contributes to this outcome?

Answer :

Option Id

(A) Routing set in a prescribed area. 412001

(B) Work centers designed for producing similar goods. 412002

(C) Mechanical transfer of materials between workstations. 412003

(D) The line setting the overall production capacity. 412004

Right Answer :

Mechanical transfer of materials between workstations.

Right Option Id : 412003

Question 38

Question Id : 413

In repetitive manufacturing, why is it important that working at one work center takes almost the same amount of time as at any other center?

Answer :

Option Id

(A) It ensures smooth flow and minimizes bottlenecks in the production line. 413001

(B) It allows the line capacity to be adjusted dynamically. 413002

(C) It reduces the need for mechanical transfer systems between workstations. 413003

(D) It increases the variety of goods that can be produced at each work center. 413004

Right Answer :

It ensures smooth flow and minimizes bottlenecks in the production line.

Right Option Id : 413001

Question 39

Question Id : 414

For a rectangular bar under axial pull, how is the average tensile normal stress across a vertical section calculated?

Answer :

Option Id

(A) Divide the internal resultant normal force by the cross-sectional area. 414001

(B) Integrate the internal normal force across the entire bar. 414002

(C) Multiply the axial force by the cross-sectional area. 414003

(D) Use the difference between the tensile and compressive forces. 414004

Right Answer :

Divide the internal resultant normal force by the cross-sectional area.

Right Option Id : 414001

Question 40

Question Id : 415

What conditions must be satisfied for the equilibrium of a bar subjected to shear forces parallel to its cross-section?

Answer :

Option Id

(A) The shear force must equal the external forces applied parallel to the cross-section. 415001

(B) The shear stress must equal the normal stress on the cross-section. 415002

(C) The resultant internal force must be perpendicular to the cross-section. 415003

(D) The average shear stress must equal the applied load divided by the cross-sectional area. 415004

Right Answer :
The shear force must equal the external forces applied parallel to the cross-section.

Right Option Id : 415001

Question 41

Question Id : 416

During a tensile test on mild steel, the stress-strain curve is linear up to point A. What is the significance of this point?

Answer :

Option Id

(A) It marks the elastic limit where the material springs back to its original shape. 416001

(B) It represents the proportional limit where stress is directly proportional to strain. 416002

(C) It signifies the upper yield point where plastic deformation begins. 416003

(D) It indicates the ultimate stress point where the material withstands maximum load. 416004

Right Answer :
It represents the proportional limit where stress is directly proportional to strain.

Right Option Id : 416002

Question 42

Question Id : 417

Why does the cross-sectional area of a mild steel bar reduce rapidly near the fracture point (Point F) during a tensile test?

Answer :

Option Id

(A) Due to uniform deformation across the bar's length. 417001

(B) Because the material undergoes necking, concentrating stress in a localized region. 417002

(C) Due to the simultaneous reduction in stress and strain after the ultimate point. 417003

(D) Because the fracture strength is greater than the ultimate strength. 417004

Right Answer :
Because the material undergoes necking concentrating stress in a localized region.

Right Option Id : 417002

Question 43

Question Id : 418

If a material is loaded beyond the yield point to a point, say point P, and then unloaded, how is the total strain at point P distributed?

Answer :

Option Id

(A) Entirely as elastic strain recovered during unloading. 418001

(B) As a sum of plastic strain recovered during unloading and elastic strain remaining in the material. 418002

(C) As a sum of elastic strain recovered during unloading and permanent plastic strain. 418003

(D) Entirely as plastic strain retained in the material after unloading. 418004

Right Answer :
As a sum of elastic strain recovered during unloading and permanent plastic strain.

Right Option Id : 418003

Question 44

Question Id : 419

If an isotropic rectangular bar experiences a lateral strain of -0.002 in its width when subjected to a tensile load, which of the following best describes the change in its transverse dimensions?

Answer :

Option Id

(A) There is no change in thickness of the bar. 419001

(B) The thickness and width reduce equally due to uniform contraction. 419002

(C) The width reduces more than the thickness. 419003

(D) The change in thickness depends on the Poisson's ratio of the material. 419004

Right Answer :
The thickness and width reduce equally due to uniform contraction.

Right Option Id : 419002

Question 45

A cylindrical bar with an initial diameter of 20 mm is subjected to a tensile force, resulting in a diametral contraction of 0.02 mm. What is the lateral strain experienced by the bar?

Answer :

Option Id

(A) -0.0005 — 420001
(B) -0.001 — 420002
(C) 0.001 — 420003
(D) -0.002 — 420004

Right Answer :

-0.001

Right Option Id : 420002

Question 46

Question Id : 421

How are standard parts, such as fasteners and gears, depicted in technical drawings?

Answer :

Option Id

(A) Shown solid black — 421001
(B) Sectioned to show interior details — 421002
(C) Drawn showing all their exterior features — 421003
(D) Lined at different angles — 421004

Right Answer :

Drawn showing all their exterior features

Right Option Id : 421003

Question 47

Question Id : 422

Why are thin features like ribs, webs, and gear teeth not section lined when the cutting plane passes parallel to them?

Answer :

Option Id

(A) To show the part as thicker — 422001
(B) To maintain the correct impression of thickness — 422002
(C) To simplify the drawing — 422003
(D) To highlight the features — 422004

Right Answer :

To maintain the correct impression of thickness

Right Option Id : 422002

Question 48

Question Id : 423

An engineer compares asynchronous and synchronous counters for a high-speed application. Why would a synchronous counter be preferred?

Answer :

Option Id

(A) Synchronous counters use fewer flip-flops than asynchronous counters. — 423001
(B) Synchronous counters allow all flip-flops to be clocked simultaneously. — 423002
(C) Synchronous counters have a lower modulus than asynchronous counters. — 423003
(D) Synchronous counters do not require clock signals. — 423004

Right Answer :

Synchronous counters allow all flip-flops to be clocked simultaneously.

Right Option Id : 423002

Question 49

Question Id : 424

In an inverting amplifier circuit, if the feedback resistor is increased while keeping the input resistor constant, what will be the resulting impact on gain, input impedance, and output voltage?

Answer :

Option Id

(A) Gain increases, input impedance remains constant, output voltage increases proportionally. — 424001
(B) Gain decreases, input impedance decreases, output voltage decreases proportionally. — 424002
(C) Gain increases, input impedance decreases, output voltage remains unaffected. — 424003

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(D) Gain remains constant, input impedance remains constant, output voltage increases.

Right Answer :

Gain increases input impedance remains constant output voltage increases proportionally.

Right Option Id : 424001

Question 50

Question Id : 425

A differentiator circuit is designed with Rf = 10kΩ and Cin = 0.1 muF. If the input signal changes at a rate of 5V/mus, what will the output voltage be?

Answer :

Option Id

(A) -5kV — 425001

(B) 5kV — 425002

(C) -10kV — 425003

(D) 10kV — 425004

Right Answer :

-5kV

Right Option Id : 425001

Question 51

Question Id : 426

A differentiator circuit receives a sine wave input. What will the output waveform look like, and why?

Answer :

Option Id

(A) A sine wave of the same phase, because the circuit blocks high-frequency components. — 426001

(B) A cosine wave, because differentiation shifts the phase of a sine wave by 90°. — 426002

(C) A square wave, because the circuit produces output proportional to input rate of change. — 426003

(D) A sine wave of inverted phase, because the circuit reverses the signal polarity. — 426004

Right Answer :

A cosine wave because differentiation shifts the phase of a sine wave by 90°.

Right Option Id : 426002

Question 52

Question Id : 427

A manufacturing process has a Process Capability Ratio (Cp) of 0.85. What does this indicate about the process, and what steps should be taken to improve it?

Answer :

Option Id

(A) The process is capable, but it needs optimization for better centering. — 427001

(B) The process is incapable, and the process variation should be brought into tolerance limits. — 427002

(C) The process is capable, but the mean needs to be aligned with the target. — 427003

(D) The process is centered but requires reducing the specification limits. — 427004

Right Answer :

The process is incapable and the process variation should be brought into tolerance limits.

Right Option Id : 427002

Question 53

Question Id : 428

At what angle are section lines typically drawn to the principal edges or axes of the part?

Answer :

Option Id

(A) 30° — 428001

(B) 45° — 428002

(C) 60° — 428003

(D) 90° — 428004

Right Answer :

45°

Right Option Id : 428002

Question 54

What is the purpose of a broken-out section in technical drawings?

Answer :

Option Id

(A) To add detail to the entire part — 429001

(B) To save time and avoid confusion — 429002

(C) To show multiple views of a part — 429003

(D) To enhance the aesthetic appeal — 429004

Right Answer :

To save time and avoid confusion

Right Option Id : 429002

Question 55

Question Id : 430

A machine operates for 120 hours in a week, during which it is idle for 20 hours. If its actual running time is 90 hours and its standard running time is 80 hours, what is its efficiency?

Answer :

Option Id

(A) 0.75 — 430001

(B) 0.8 — 430002

(C) 0.8889 — 430003

(D) None of these — 430004

Right Answer :

0.8889

Right Option Id : 430003

Question 56

Question Id : 431

A production line employs passive inspection but experiences high scrap rates due to defects. Which change is most effective in improving defect detection and reducing scrap?

Answer :

Option Id

(A) Switch to active inspection to monitor components during production. — 431001

(B) Increase the frequency of passive inspections throughout the production line. — 431002

(C) Modify the inspection process to include more precise tools for measuring defects. — 431003

(D) None of these — 431004

Right Answer :

Switch to active inspection to monitor components during production.

Right Option Id : 431001

Question 57

Question Id : 432

A lathe machine produces shafts with a target mean width of 25.0 mm and a standard deviation of 0.005 mm. If the process is stable, what percentage of shafts are expected to have widths within the range of 24.985 mm to 25.015 mm?

Answer :

Option Id

(A) 68.27% — 432001

(B) 95.45% — 432002

(C) 99.73% — 432003

(D) None of these — 432004

Right Answer :

99.73%

Right Option Id : 432003

Question 58

Question Id : 433

A company aims to achieve a target profit of Rs.300,000. If fixed costs are Rs.400,000, the variable cost per unit is Rs.12, and the price per unit is Rs.20, what is the minimum sales volume required?

Answer :

Option Id

(A) 50,000 units — 433001

(B) 70,000 units — 433002

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(C) 87,500 units
(D) 100,000 units

433004

Right Answer :
50 000 units

Right Option Id : 433001

Question 59

Question Id : 434

In a double-shear joint, a rivet with a cross-sectional area of 80 mm^2 per shear plane is subjected to a load of 16,000 N. What is the shear stress induced in the rivet?

Answer : Option Id

(A) 50 MPa 434001
(B) 100 MPa 434002
(C) 1.25 MPa 434003
(D) 2 MPa 434004

Right Answer :
100 MPa

Right Option Id : 434002

Question 60

Question Id : 435

What is the result of connecting the points where the lines of sight pierce the projection plane?

Answer : Option Id

(A) The 3D object is rotated 435001
(B) The 3D object is mirrored 435002
(C) The 3D object is transformed into a 2D view 435003
(D) The 3D object is scaled 435004

Right Answer :
The 3D object is transformed into a 2D view

Right Option Id : 435003

Question 61

Question Id : 436

In which type of projection are the object surface and the projection plane parallel, but the lines of sight are not perpendicular to the projection plane?

Answer : Option Id

(A) Multiview orthographic projection 436001
(B) Isometric view (orthographic) 436002
(C) Oblique projection (non-orthographic) 436003
(D) Perspective projection 436004

Right Answer :
Oblique projection (non-orthographic)

Right Option Id : 436003

Question 62

Question Id : 437

For three specimens of the same material with different cross-sectional areas and lengths, what must be normalized to generate a unified stress-strain curve?

Answer : Option Id

(A) Load by cross-sectional area and deformation by original length. 437001
(B) Load by original length and deformation by cross-sectional area. 437002
(C) Deformation by cross-sectional area and load by applied force. 437003
(D) Load by deformation and cross-sectional area by original length. 437004

Right Answer :
Load by cross-sectional area and deformation by original length.

Right Option Id : 437001

Question 63

Question Id : 438

If the length of an axially loaded bar is halved and its cross-sectional area is doubled, how will its stiffness and flexibility change?

Answer :

Option Id

(A) Stiffness doubles, and flexibility remains unchanged. 438001

(B) Stiffness increases by four times, and flexibility decreases by four times. 438002

(C) Stiffness doubles, and flexibility also doubles. 438003

(D) Stiffness remains constant, and flexibility decreases by half. 438004

Right Answer :

Stiffness increases by four times and flexibility decreases by four times.

Right Option Id : 438002

Question 64

Question Id : 439

How does the manner of load application affect the stress distribution in a bar according to St. Venant's Principle?

Answer :

Option Id

(A) It changes the stress distribution throughout the bar. 439001

(B) It only affects stress distribution near the points of load application. 439002

(C) It has no impact as long as the resultant load remains the same. 439003

(D) It causes stress variations at both ends of the bar. 439004

Right Answer :

It only affects stress distribution near the points of load application.

Right Option Id : 439002

Question 65

Question Id : 440

If a structural component has a yield stress of 400 MPa and is designed with a factor of safety of 2.5, what is the allowable working stress for the component?

Answer :

Option Id

(A) 0.625 MPa 440001

(B) 160 MPa 440002

(C) 240 MPa 440003

(D) None of these 440004

Right Answer :

160 MPa

Right Option Id : 440002

Question 66

Question Id : 441

If a symmetric rectangular beam is subjected to a negative bending moment, where will the maximum tensile stress occur, and why?

Answer :

Option Id

(A) At the neutral axis, because stress is zero there. 441001

(B) At the bottom fibre, because the fibres below the neutral axis are in tension. 441002

(C) At the top fibre, because the fibres above the neutral axis experience tensile stress in negative bending. 441003

(D) At the midpoint of the beam's length, because bending is maximum there. 441004

Right Answer :

At the top fibre because the fibres above the neutral axis experience tensile stress in negative bending.

Right Option Id : 441003

Question 67

Question Id : 442

What is the basis of multi-view projection in technical drawing?

Answer :

Option Id

(A) Perspective projection 442001

(B) Isometric projection 442002

(C) Orthographic (parallel) projection 442003

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(D) Oblique projection

Right Answer :
Orthographic (parallel) projection

Right Option Id : 442003

Question 68

Question Id : 443

Which thermodynamic process occurs at constant temperature?

Answer :

Option Id

(A) Adiabatic — 443001
(B) Isobaric — 443002
(C) Isochoric — 443003
(D) Isothermal — 443004

Right Answer :
Isothermal

Right Option Id : 443004

Question 69

Question Id : 444

In an automated process, how does a feedback control system contribute to maintaining operational accuracy?

Answer :

Option Id

(A) It directly executes the instructions from the plan without any corrections. — 444001
(B) It modifies the power source to ensure consistent output. — 444002
(C) It continuously monitors and adjusts operations based on system performance. — 444003
(D) It ensures the system operates continuously without interruptions regardless of errors. — 444004

Right Answer :
It continuously monitors and adjusts operations based on system performance.

Right Option Id : 444003

Question 70

Question Id : 445

How does enhanced flexibility improve automated systems' adaptability in a manufacturing environment?

Answer :

Option Id

(A) By allowing heavy materials to be handled more safely using automated systems. — 445001
(B) By enabling real-time adjustments to production configurations and processes. — 445002
(C) By eliminating standalone manufacturing machines for specific tasks. — 445003
(D) By increasing the number of operations performed simultaneously on a single workpiece. — 445004

Right Answer :
By enabling real-time adjustments to production configurations and processes.

Right Option Id : 445002

Question 71

Question Id : 446

Which of the following is a primary advantage of numerical control machines compared to conventional machines?

Answer :

Option Id

(A) Ability to adjust cutting speed dynamically during operations. — 446001
(B) Flexibility to adapt to design changes without altering production equipment. — 446002
(C) Low initial investment and maintenance costs. — 446003
(D) Elimination of skilled labor for operation and maintenance. — 446004

Right Answer :
Flexibility to adapt to design changes without altering production equipment.

Right Option Id : 446002

Question 72

Which code is responsible for controlling operations unrelated to movement, such as turning the spindle on or off, in a CNC machine?

Answer :

Option Id

(A) G-code — 447001
(B) M-code — 447002
(C) F-code — 447003
(D) D-code — 447004

Right Answer :
M-code

Right Option Id : 447002

Question 73

Question Id : 448

What activates the CNC machine and its logic controller to autonomously create a part?

Answer :

Option Id

(A) The machine control unit — 448001
(B) G and M codes — 448002
(C) Toolpath movement instructions — 448003
(D) Cycle start button — 448004

Right Answer :
Cycle start button

Right Option Id : 448004

Question 74

Question Id : 449

Which of the following processes is primarily used for shaping metal into desired forms using a die?

Answer :

Option Id

(A) Casting — 449001
(B) Welding — 449002
(C) Rolling — 449003
(D) Extrusion — 449004

Right Answer :
Extrusion

Right Option Id : 449004

Question 75

Question Id : 450

Which type of internal combustion engine operates on the principle of ignition by compression?

Answer :

Option Id

(A) Diesel Engine — 450001
(B) Petrol Engine — 450002
(C) Gasoline Engine — 450003
(D) Hydrogen Engine — 450004

Right Answer :
Diesel Engine

Right Option Id : 450001

Question 76

Question Id : 451

Why are semi-synthetic fluids considered a balanced option for machining operations?

Answer :

Option Id

(A) They combine synthetic fluids and soluble oils, offering mid-range cooling and lubrication. — 451001
(B) They contain corrosion inhibitors and are entirely water-based for maximum cooling performance. — 451002
(C) They consist of only 5-20% mineral oil, providing effective cooling at a moderate price. — 451003
(D) They are entirely biodegradable and environmentally friendly, ideal for all machining operations. — 451004

Right Answer :
They consist of only 5-20% mineral oil providing effective cooling at a moderate price.

Right Option Id : 451003

Question 77

Question Id : 452

What unique feature of the compound rest allows precise angular adjustments during taper turning operations?

Answer :

Option Id

(A) The tongue and groove joint — 452001
(B) The dove-tail joint — 452002
(C) The circular base with angular graduations — 452003
(D) The mounting holes — 452004

Right Answer :
The circular base with angular graduations

Right Option Id : 452003

Question 78

Question Id : 453

Why is fixed automation considered less suitable for products with frequent design changes?

Answer :

Option Id

(A) It lacks the flexibility to modify equipment settings efficiently. — 453001
(B) It requires specialized equipment that cannot adapt to varying production requirements. — 453002
(C) Its high initial cost outweighs the benefits for low-volume production. — 453003
(D) It integrates multiple processes, making modifications costly and time-consuming. — 453004

Right Answer :
It integrates multiple processes making modifications costly and time-consuming.

Right Option Id : 453004

Question 79

Question Id : 454

Why is the changeover process in programmable automation considered both an advantage and a limitation?

Answer :

Option Id

(A) It requires minimal programming but demands extensive physical setup. — 454001
(B) It allows product customization but necessitates reprogramming and physical adjustments. — 454002
(C) It supports high-volume production but restricts flexibility in product configurations. — 454003
(D) It enables seamless manufacturing but limits the use of advanced robotics. — 454004

Right Answer :
It allows product customization but necessitates reprogramming and physical adjustments.

Right Option Id : 454002

Question 80

Question Id : 455

What is a key advantage of flexible automation in managing production of varied products?

Answer :

Option Id

(A) It reduces human involvement in monitoring and maintenance. — 455001
(B) It relies exclusively on batch processing for efficiency. — 455002
(C) It allows seamless transitions between different products without grouping them into batches. — 455003
(D) It eliminates the need for programmable controllers and CNC machines. — 455004

Right Answer :
It allows seamless transitions between different products without grouping them into batches.

Right Option Id : 455003

Question 81

Question Id : 456

What type of stress does a twisting force (torque) induce in a circular shaft?

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Answer :

(A) Shear stress — 456001
(B) Tensile stress — 456002
(C) Compressive stress — 456003
(D) Bending stress — 456004

Right Answer :
Shear stress

Right Option Id : 456001

Question 82

Question Id : 457

Which of the following is an example of a non-ferrous metal?

Answer : Option Id

(A) Brass — 457001
(B) Cast Iron — 457002
(C) Steel — 457003
(D) Carbon Steel — 457004

Right Answer :
Brass

Right Option Id : 457001

Question 83

Question Id : 458

How does CNC technology enhance production efficiency compared to manual operations?

Answer : Option Id

(A) By eliminating human intervention entirely from the manufacturing process. — 458001
(B) By automating design iterations without any programming updates. — 458002
(C) By improving precision and enabling continuous operation for longer periods. — 458003
(D) By allowing real-time adjustments to workpiece dimensions during machining. — 458004

Right Answer :
By improving precision and enabling continuous operation for longer periods.

Right Option Id : 458003

Question 84

Question Id : 459

What critical dependency must be ensured when using CAM software for CNC programming to manufacture a part with intricate shapes and dimensions?

Answer : Option Id

(A) The G-code must be manually input into the CNC machine for precise control. — 459001
(B) The part's 3D model must be available to generate tool paths and subsequent G-code. — 459002
(C) The machine control unit must be calibrated to interpret M-code instructions. — 459003
(D) The cutter diameter and feed rate must be pre-determined and manually adjusted. — 459004

Right Answer :
The part's 3D model must be available to generate tool paths and subsequent G-code.

Right Option Id : 459002

Question 85

Question Id : 460

Which component of the CNC machine interprets G and M codes to coordinate axis motion, and how does it manage auxiliary functions such as tool changes or coolant flow?

Answer : Option Id

(A) The driving unit processes the G and M codes to amplify signals for precise axis movements while controlling auxiliary functions. — 460001
(B) The input device interprets the G and M codes and directly sends commands for axis motion and auxiliary control to the machine control unit. — 460002
(C) The machine control unit decodes the G and M codes, coordinates axis motion, and sends signals to handle auxiliary functions. — 460003
(D) The tools unit directly reads G and M codes to set the spindle speed and execute motion commands while managing auxiliary operations. — 460004

Right Answer :
The machine control unit decodes the G and M codes coordinates axis motion and sends signals to handle auxiliary functions.

Right Option Id : 460003

Question 86

Question Id : 461

Which laser type is commonly used in CNC laser cutting, and why is it effective for cutting metal sheets?

Answer :

Option Id

(A) Carbon dioxide lasers, due to their high power and capability to cut metals precisely. 461001
(B) Plasma torches, as they offer the highest efficiency for tougher materials. 461002
(C) Nd lasers, as they are specifically designed for cutting metals at high speed. 461003
(D) YAG lasers, because they generate heat suitable only for softer materials. 461004

Right Answer :
Carbon dioxide lasers due to their high power and capability to cut metals precisely.

Right Option Id : 461001

Question 87

Question Id : 462

Why is the EDM process ideal for manufacturing components requiring high sensitivity?

Answer :

Option Id

(A) It uses mechanical forces to achieve precise dimensions on delicate materials. 462001
(B) It requires minimal programming and uses manual adjustments for accurate results. 462002
(C) It employs high-pressure coolant systems to stabilize machining operations. 462003
(D) It avoids cutting forces entirely, relying on electrical discharge to remove material. 462004

Right Answer :
It avoids cutting forces entirely relying on electrical discharge to remove material.

Right Option Id : 462004

Question 88

Question Id : 463

What is the ideal air-fuel ratio for a petrol engine under normal operating conditions?

Answer :

Option Id

(A) 10:1 463001
(B) 14.7:1 463002
(C) 16:1 463003
(D) 20:1 463004

Right Answer :
14.7:1

Right Option Id : 463002

Question 89

Question Id : 464

Which of the following describes the condition when the velocity of flow is the same at any given instant?

Answer :

Option Id

(A) Turbulent flow 464001
(B) Unsteady flow 464002
(C) Rotational flow 464003
(D) Uniform flow 464004

Right Answer :
Uniform flow

Right Option Id : 464004

Question 90

Question Id : 465

Which feature of 5-axis CNC machining contributes to its ability to produce intricate structures with minimal setup?

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Answer :

(A) Increased spindle speed allowing faster material removal rates. 465001
(B) Integration of additive and subtractive processes to manufacture parts. 465002
(C) Simultaneous movement along three linear and two rotational axes to machine complex geometries. 465003
(D) Automated tool-changing systems enabling continuous operations without manual intervention. 465004

Right Answer :
Simultaneous movement along three linear and two rotational axes to machine complex geometries.

Right Option Id : 465003

Question 91

Question Id : 466

Which factor makes continuous work-part transport less suitable for processing heavy workpieces?

Answer : Option Id

(A) Limited speed of the conveyor system. 466001
(B) Inertial challenges caused by the weight and size of work-heads. 466002
(C) High maintenance requirements of conveyor systems for heavy loads. 466003
(D) Difficulty in achieving precise alignment during processing. 466004

Right Answer :
Inertial challenges caused by the weight and size of work-heads.

Right Option Id : 466002

Question 92

Question Id : 467

Which fluid is preferred for machining cast iron materials, and what concentration range is recommended for the optimal performance?

Answer : Option Id

(A) Emulsions provide superior cooling and surface finish; concentration should be 10-15%. 467001
(B) Cutting oils offer better lubrication; concentration should be 5-10%. 467002
(C) Emulsions prevent chip adhesion; concentration should be 15-20%. 467003
(D) Cutting oils reduce friction; concentration should be 10-15%. 467004

Right Answer :
Emulsions provide superior cooling and surface finish; concentration should be 10-15%.

Right Option Id : 467001

Question 93

Question Id : 468

Which innovation by Henry Maudslay transformed the lathe into a tool capable of metal cutting, and how did it impact machining accuracy?

Answer : Option Id

(A) Development of the motorized lathe; enabled higher cutting speeds. 468001
(B) Use of the operator's foot for motion; enhanced control over cutting. 468002
(C) Invention of springy branches for motion; stabilized cutting processes. 468003
(D) Introduction of the sliding carriage; provided precise tool guidance. 468004

Right Answer :
Introduction of the sliding carriage; provided precise tool guidance.

Right Option Id : 468004

Question 94

Question Id : 469

Which feature distinguishes a CNC lathe from a geared lathe in terms of operational precision and versatility?

Answer : Option Id

(A) The use of a lead screw and feed shaft for increased machining capacity. 469001
(B) The ability to perform shaping tasks using manual controls. 469002
(C) Digital motors enabling multi-axis movement with precise design instructions. 469003
(D) The integration of heavy-duty capabilities for processing tough materials. 469004

Right Answer :
Digital motors enabling multi-axis movement with precise design instructions.

Right Option Id : 469003

Question 95

Question Id : 470

What is the term for the resistance of a material to indentation or scratching?

Answer :

Option Id

(A) Hardness — 470001
(B) Toughness — 470002
(C) Strength — 470003
(D) Ductility — 470004

Right Answer :
Hardness

Right Option Id : 470001

Question 96

Question Id : 471

Which material is primarily used for manufacturing engine blocks in vehicles?

Answer :

Option Id

(A) Nickel — 471001
(B) Copper — 471002
(C) Cast Iron — 471003
(D) Steel — 471004

Right Answer :
Cast Iron

Right Option Id : 471003

Question 97

Question Id : 472

What critical role do the guideways play in ensuring the precision and stability of a lathe machine's operation?

Answer :

Option Id

(A) They act as the primary support for the headstock and tailstock. — 472001
(B) They absorb vibrations caused by cutting forces to maintain stability. — 472002
(C) They facilitate the movement of the carriage and tool holder along a pre-defined path. — 472003
(D) They provide a horizontal foundation for all lathe components. — 472004

Right Answer :
They facilitate the movement of the carriage and tool holder along a pre-defined path.

Right Option Id : 472003

Question 98

Question Id : 473

How does the carriage ensure both manual and automatic movement along the lathe bed?

Answer :

Option Id

(A) By housing the saddle that supports cross-slide movements. — 473001
(B) By incorporating a lead screw for automatic threading operations. — 473002
(C) By enabling gear adjustments in the apron for feed mode control. — 473003
(D) By maintaining alignment with the guideways for precise motion. — 473004

Right Answer :
By incorporating a lead screw for automatic threading operations.

Right Option Id : 473002

Question 99

Question Id : 474

What specific role does the saddle play within the carriage assembly of a lathe machine?

Answer :

Option Id

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(A) It houses the compound rest for angular tool adjustments.
(B) It includes the lead screw for automated threading operations.
(C) It supports the cross slide and provides stability for tool post movements.
(D) It incorporates the gear box for controlling feed direction and speed.

474001
474002
474003
474004

Right Answer :
It supports the cross slide and provides stability for tool post movements.

Right Option Id : 474003

Question 100

Question Id : 475

What primary advantage does a four-way tool post offer over a single way tool post in machining operations?
Answer :

(A) It secures the tools with a single locking screw for greater stability.
(B) It allows quick switching between tools for increased machining speed.
(C) It uses a concave ring and convex rocker for precise tool alignment.
(D) It eliminates the need for a locking bolt by clamping tools directly.

Option Id
475001
475002
475003
475004

Right Answer :
It allows quick switching between tools for increased machining speed.

Right Option Id : 475002

Question 101

Question Id : 476

How does the engagement of the half nut lever with the lead screw ensure precise carriage movement during thread cutting?
Answer :

(A) By using the Acme thread design to resist axial loads during movement.
(B) By locking the carriage to the lead screw to synchronize movement with threading.
(C) By transmitting rotational motion directly to the tool post.
(D) By guiding the carriage along the guideways for uniform thread depth.

Option Id
476001
476002
476003
476004

Right Answer :
By locking the carriage to the lead screw to synchronize movement with threading.

Right Option Id : 476002

Question 102

Question Id : 477

What is meant by a "no-slip" condition in CFD?
Answer :

(A) No velocity difference between the fluid and the surface
(B) No pressure difference at the boundary
(C) No turbulence near the boundary
(D) No heat exchange between the fluid and the surface

Option Id
477001
477002
477003
477004

Right Answer :
No velocity difference between the fluid and the surface

Right Option Id : 477001

Question 103

Question Id : 478

What does the term “robot kinematics“ refer to?

Answer :

Option Id

(A) The movement of robots in a virtual environment — 478001

(B) The study of the motion of robot parts — 478002

(C) The process of robot programming — 478003

(D) The design of robot control systems — 478004

Right Answer :

The study of the motion of robot parts

Right Option Id : 478002

Question 104

Question Id : 479

What primary design characteristic of lathe legs supports their ability to handle the tool’s weight and maintain stability?

Answer :

Option Id

(A) Commercial-grade lathes have legs with foundation bolts for floor attachment. — 479001

(B) The legs are elevated to create an ergonomic working surface. — 479002

(C) Legs are designed with an Acme thread structure for load distribution. — 479003

(D) The vertical components of legs allow precise carriage alignment during machining. — 479004

Right Answer :

Commercial-grade lathes have legs with foundation bolts for floor attachment.

Right Option Id : 479001

Question 105

Question Id : 480

In a compressible flow through a convergent-divergent nozzle, when the back pressure decreases below the critical pressure, what will be the mass flow rate?

Answer :

Option Id

(A) It will increase — 480001

(B) It will decrease — 480002

(C) It will first increase then decrease — 480003

(D) It will remain constant — 480004

Right Answer :

It will remain constant

Right Option Id : 480004

Question 106

Question Id : 481

Which of the following is the true purpose of the feed rod in the machine?

Answer :

Option Id

(A) Feeding rod through the hollow spindle in lathes with chucks — 481001

(B) Automatic feed — 481002

(C) Cutting multiple threads — 481003

(D) Taper turning — 481004

Right Answer :

Automatic feed

Right Option Id : 481002

Question 107

Question Id : 482

Which of the following is not correct for the statement relevant to the streamlines?

Answer :

Option Id

(A) The orientation will be fixed in a steady flow — 482001

(B) Given by eulerian method — 482002

(C) Group of the streamlines for a flow- field is known as steam-tube — 482003

(D) Locus of the temporary locations of all particles that have passed through a fixed point — 482004

Right Answer :
Locus of the temporary locations of all particles that have passed through a fixed point

Right Option Id : 482004

Question 108

Question Id : 483

In a pipe flow, the head loss due to friction is 5 m. If the power transmitted through the pipe has to be the maximum, then the total head at the inlet of the pipe will have to be maintained at _____.

Answer :

Option Id

(A) 10 m — 483001
(B) 15 m — 483002
(C) 20 m — 483003
(D) None of these — 483004

Right Answer :
15 m

Right Option Id : 483002

Question 109

Question Id : 484

What is the term for reducing waste at the source in sustainable manufacturing?

Answer :

Option Id

(A) Recycling — 484001
(B) Pollution control — 484002
(C) Waste minimization — 484003
(D) Process optimization — 484004

Right Answer :
Waste minimization

Right Option Id : 484003

Question 110

Question Id : 485

Which type of vibration occurs when a system oscillates in its natural mode without any external force?

Answer :

Option Id

(A) Forced vibration — 485001
(B) Complex vibration — 485002
(C) Damped vibration — 485003
(D) Free vibration — 485004

Right Answer :
Free vibration

Right Option Id : 485004

Question 111

Question Id : 486

Which thermodynamics law gives the steady flow energy equation for turbomachines?

Answer :

Option Id

(A) First law of thermodynamics — 486001
(B) Zeroth law of thermodynamics — 486002
(C) Second law of thermodynamics — 486003
(D) Third law of thermodynamics — 486004

Right Answer :
First law of thermodynamics

Right Option Id : 486001

Question 112

Question Id : 487

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In the infinite Prandtl number limit, why is the velocity field said to be "slaved" by the temperature field?

Answer :

Option Id

(A) Because the velocity field dominates the temperature diffusion process. 487001

(B) Because temperature diffusion occurs more rapidly than velocity diffusion. 487002

(C) Because the velocity field diffuses more rapidly than the temperature field. 487003

(D) Because the velocity field remains independent of the temperature field. 487004

Right Answer :

Because the velocity field diffuses more rapidly than the temperature field.

Right Option Id : 487003

Question 113

Question Id : 488

The direct-contact heat exchanger used in an actual vapour absorption refrigeration plant for elimination of water vapour from the ammonia vapour before entering condenser is known as ___.

Answer :

Option Id

(A) Rectifier 488001

(B) Absorber 488002

(C) Condenser 488003

(D) Analyzer 488004

Right Answer :

Analyzer

Right Option Id : 488004

Question 114

Question Id : 489

A sphere with a diameter of 0.1 m and a temperature of 150°C is immersed in water at 20°C. The convective heat transfer coefficient between the sphere surface and the water is 500 W/m^2K. What is the rate of heat transfer from the sphere to the water?

Answer :

Option Id

(A) 1.950 kW 489001

(B) 2.005 kW 489002

(C) 2.041 kW 489003

(D) 2.096 kW 489004

Right Answer :

2.041 kW

Right Option Id : 489003

Question 115

Question Id : 490

Which type of engine cycle involves intake, compression, power, and exhaust strokes?

Answer :

Option Id

(A) Two-stroke cycle 490001

(B) Diesel cycle 490002

(C) Four-stroke cycle 490003

(D) Brayton cycle 490004

Right Answer :

Four-stroke cycle

Right Option Id : 490003

Question 116

Question Id : 491

Which sensor is commonly used in robotics for detecting obstacles?

Answer :

Option Id

(A) Ultrasonic Sensor 491001

(B) Temperature Sensor 491002

(C) Humidity Sensor 491003

(D) Light Sensor 491004

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Right Answer :
Ultrasonic Sensor

Right Option Id : 491001

Question 117

Question Id : 492

A mass of 5 kg is attached to a spring with a spring constant of 1000 N/m. The mass is set in motion with an initial velocity of 2 m/s, and the amplitude of the motion is 0.1 m. What is the maximum acceleration of the mass?

Answer :

Option Id

(A) 10 m/s^2 — 492001
(B) 20 m/s^2 — 492002
(C) 30 m/s^2 — 492003
(D) 40 m/s^2 — 492004

Right Answer :
20 m/s^2

Right Option Id : 492002

Question 118

Question Id : 493

A solid circular steel shaft (shear modulus = 90 GPa) 1.5 m long of cross-sectional area 100 mm^2 is used for transmitting power from one pulley to another. What is the maximum strain energy stored for maximum allowable stress 60 MPa?

Answer :

Option Id

(A) 15,000 Nmm — 493001
(B) 1,500 Nmm — 493002
(C) 3,000 Nmm — 493003
(D) 30,000 Nmm — 493004

Right Answer :
3 000 Nmm

Right Option Id : 493003

Question 119

Question Id : 494

Which sensor type is most commonly used to detect the presence of objects in a mechatronic system?

Answer :

Option Id

(A) Capacitive sensor — 494001
(B) Force sensor — 494002
(C) Gyroscope — 494003
(D) Proximity sensor — 494004

Right Answer :
Proximity sensor

Right Option Id : 494004

Question 120

Question Id : 495

Which microcontroller is known for being used in Arduino projects?

Answer :

Option Id

(A) STM32 — 495001
(B) ATmega328 — 495002
(C) PIC16F877A — 495003
(D) ESP32 — 495004

Right Answer :
ATmega328

Right Option Id : 495002

Question 121

Question Id : 496

Which type of drill requires manual rotation of a wheel to spin the connected bit and is best suited for drilling softwood, plastic, and soft metal?

Answer :

Option Id

(A) Brace drill — 496001
(B) Hand drill — 496002
(C) Power drill — 496003
(D) Impact drill — 496004

Right Answer :

Hand drill

Right Option Id : 496002

Question 122

Question Id : 497

Which feature is typically machined using internal broaching?

Answer :

Option Id

(A) External threads — 497001
(B) Helical slots — 497002
(C) Keyways in through holes — 497003
(D) External surfaces — 497004

Right Answer :

Keyways in through holes

Right Option Id : 497003

Question 123

Question Id : 498

Which sensor is commonly used for measuring angular displacements and velocities?

Answer :

Option Id

(A) Accelerometer — 498001
(B) Shaft Encoders — 498002
(C) Thermocouple — 498003
(D) Pressure sensor — 498004

Right Answer :

Shaft Encoders

Right Option Id : 498002

Question 124

Question Id : 499

Which lathe operation involves using a cutting tool rotating at an angle to the workpiece's axis, and what type of surface does it create?

Answer :

Option Id

(A) Step turning; creates a series of steps with varying diameters. — 499001
(B) Boring; extends an existing hole through a specific distance. — 499002
(C) Taper Turning; produces a surface with a uniformly changing diameter along the workpiece. — 499003
(D) Threading; creates threads on a cylindrical or conical surface. — 499004

Right Answer :

Taper Turning; produces a surface with a uniformly changing diameter along the workpiece.

Right Option Id : 499003

Question 125

Question Id : 500

Which lathe operation involves modifying an existing hole and how does it achieve material removal?

Answer :

Option Id

(A) Taper Turning; removes material by creating a tapered surface. — 500001
(B) Drilling; removes material by feeding a drill bit into the rotating workpiece. — 500002
(C) Boring; removes material using a boring bar for deep cuts. — 500003

(D) Step Turning; removes material by creating multiple stepped diameters.

Right Answer :

Boring; removes material using a boring bar for deep cuts.

Right Option Id : 500003